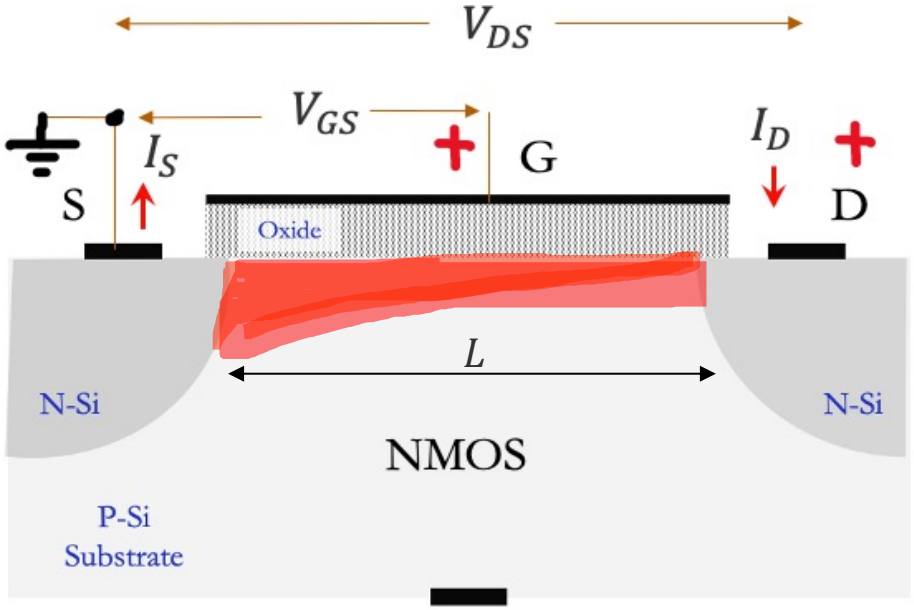
Field Effect Transistors

- Current in MOSFETs and Operation Modes.
- DC Analysis (Biasing).

Current in a MOSFET



Electron mobility — Oxide capacitance

NMOS $$k_n = \frac{\mu_n C_{ox}}{2}\frac{W}{L}$$ → Aspect ratio (size)

PMOS $$k_p = \frac{\mu_p C_{ox}}{2}\frac{W}{L}$$

1) Cutoff Mode

$$V_{GS} < V_t$$

$$I_D = 0$$

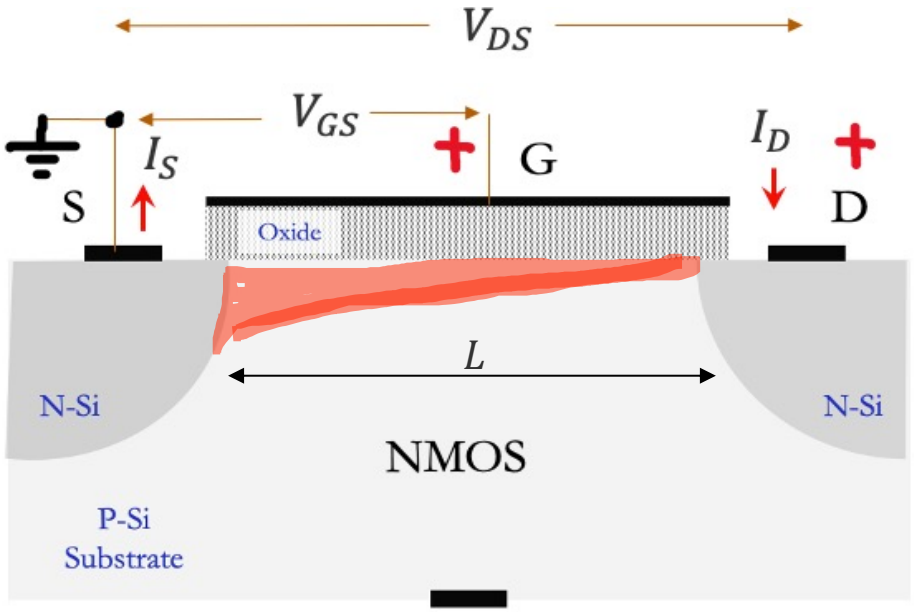
2) Triode Mode

$$V_{GS} \geq V_t \text{ and } V_{DS} \leq V_{GS} - V_t$$

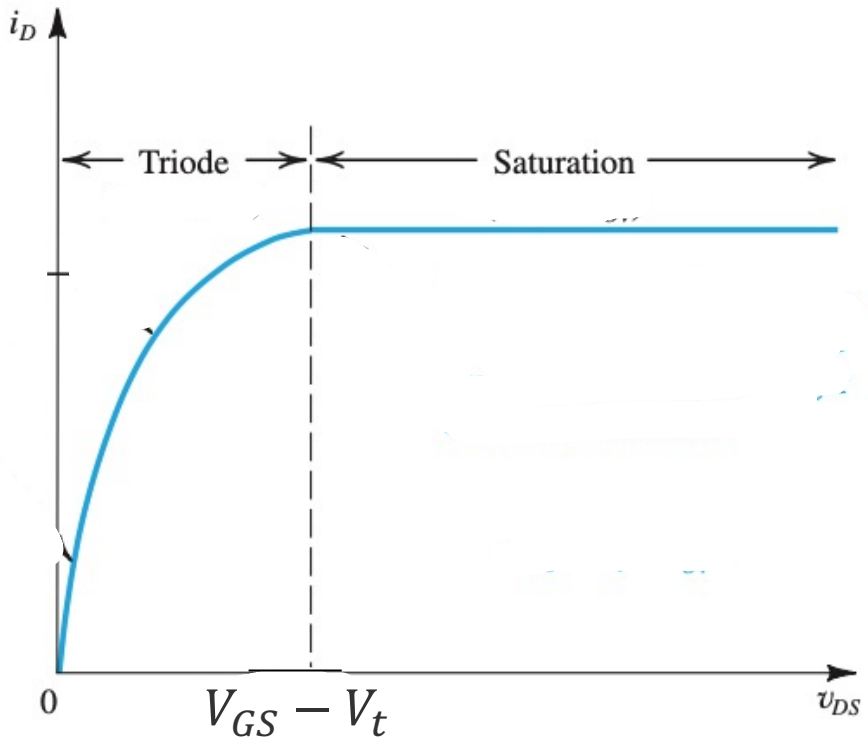
$$I_D = k\left[2(V_{GS} - V_t)V_{DS} - {V_{DS}}^2\right]$$

$$k = \frac{\mu\, C_{ox}}{2}\frac{W}{L}$$

Current in a MOSFET



MOSFET IV Curve



Pinch-off

3) Saturation Mode

$$V_{GS} \geq V_t \text{ and } V_{DS} \geq V_{GS} - V_t$$

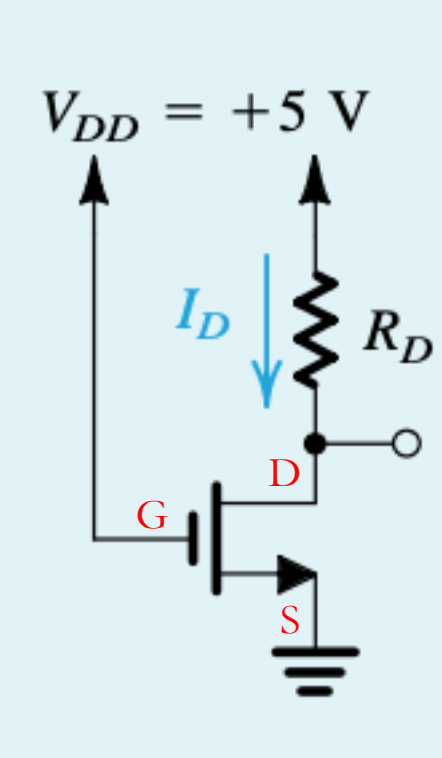
$$I_D = k(V_{GS} - V_t)^2$$

Summary

Mode	NMOS (V_t +ve)	PMOS (V_t -ve)
Cutoff	$V_{GS} < V_t \quad I_D = 0$	$V_{GS} > V_t \quad I_D = 0$
Triode	$V_{GS} \geq V_t$ and $V_{DS} \leq V_{GS} - V_t$ $I_D = k\left[2(V_{GS} - V_t)V_{DS} - {V_{DS}}^2\right]$	$V_{GS} \leq V_t$ and $V_{DS} \geq V_{GS} - V_t$ $I_D = k\left[2(V_{GS} - V_t)V_{DS} - {V_{DS}}^2\right]$
Saturation	$V_{GS} \geq V_t$ and $V_{DS} \geq V_{GS} - V_t$ $I_D = k(V_{GS} - V_t)^2$	$V_{GS} \leq V_t$ and $V_{DS} \leq V_{GS} - V_t$ $I_D = k(V_{GS} - V_t)^2$

Examples

1) Check the transistor operation mode in the circuit below $R_D = 12\ k\Omega, I_D = 0.4\ mA, V_t = 1\ V$.



$$V_S = 0$$

$$V_G = 5\ V$$

$$V_{GS} = 5 - 0 = 5\ V$$

$$V_{GS} > V_t$$

Not Cut-off

$$V_D = 5 - (12k)(0.4\ m) = 0.2\ V$$

$$V_{DS} = 0.2 - 0 = 0.2\ V$$

Check

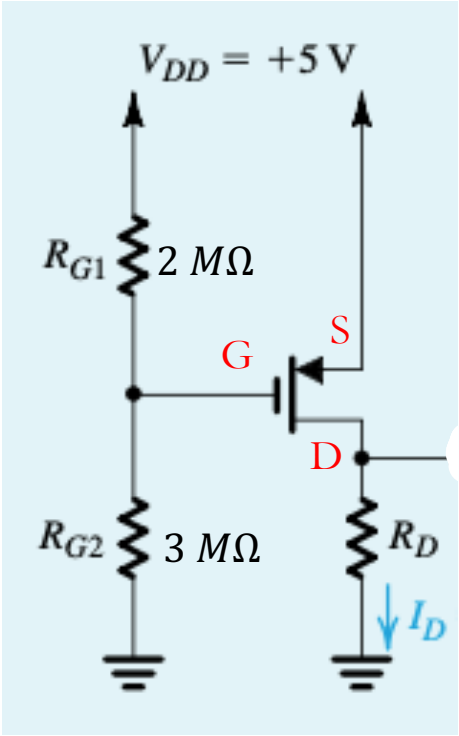
$$V_{DS} \quad ? \quad V_{GS} - V_t$$

$$0.2 \quad < \quad 5 - 1$$

Triode Mode

Examples

2) Check the transistor operation mode in the circuit below $R_D = 6\ k\Omega, I_D = 0.5\ mA, |V_t| = 1\ V$.



$$V_S = 5\ V \qquad V_G = 5\frac{3}{5} = 3\ V$$

$$V_{GS} = 3 - 5 = -2\ V \qquad V_{GS} < V_t$$

Not Cut-off

$$V_D = (6k)(0.5\ m) = 3\ V$$

$$V_{DS} = 3 - 5 = -2\ V$$

Check $\quad V_{DS} \quad ? \quad V_{GS} - V_t$

$$-2 \quad < \quad -2 - (-1)$$

Saturation Mode

$$V_D - V_S \quad ? \quad V_G - V_S - V_t$$

$$3 \quad < \quad 3 - (-1)$$

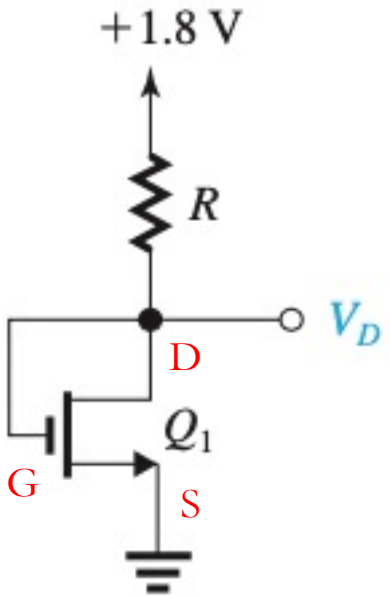
What should R_D be for edge of saturation?

$$V_D = V_G - V_t = 4\ V$$

$$R_D = \frac{4}{0.5m} = 8\ k\Omega$$

Examples

3) Analyze the transistor circuit below. Take $R_D = 13.9\ k\Omega, V_t = 0.5\ V$.



$\mu_n C_{ox} = 0.4\ \text{mA/V}^2$

$\frac{W}{L} = \frac{0.72\ \mu m}{0.18\ \mu m}$

$k_n = 0.8\ \text{mA/V}^2$

Check mode of operation

$V_S = 0 \qquad V_G = V_D$

$V_{DS} \quad ? \quad V_{GS} - V_t$

$V_D \quad > \quad V_G - V_t$

Saturation Mode

$I_D = k(V_{GS} - V_t)^2$

$I_D = 0.8\ m\ (V_G - 0.5)^2 = \frac{1.8 - V_G}{13.9\ k}$

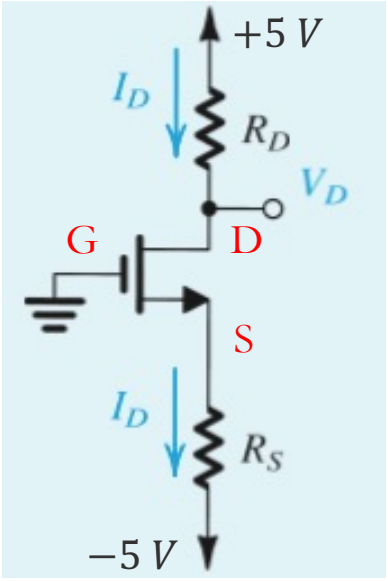
Solve for V_G

$V_G = 0.8\ V = V_D$

$I_D = 0.072\ mA$

Examples

4) Design the transistor circuit below for $I_D = 0.4\,mA, V_D = +1\,V$. Take $V_t = 2\,V$.



Check mode of operation

$$V_G = 0$$

$$V_{DS} \quad ? \quad V_{GS} - V_t$$

$$V_D \quad ? \quad V_G - V_t$$

$$+1 \quad > 0 - 2$$

Saturation Mode

$$I_D = k(V_{GS} - V_t)^2$$

$$0.4\,m = 0.4\,m\,(V_{GS} - 2)^2$$

Solve for V_{GS}

$V_{GS} = +1\,V$ Cutoff ✗

$V_{GS} = +3\,V$ ✓

$V_S = -3\,V$

$\mu_n C_{ox} = 20\ \mu\text{A/V}^2$

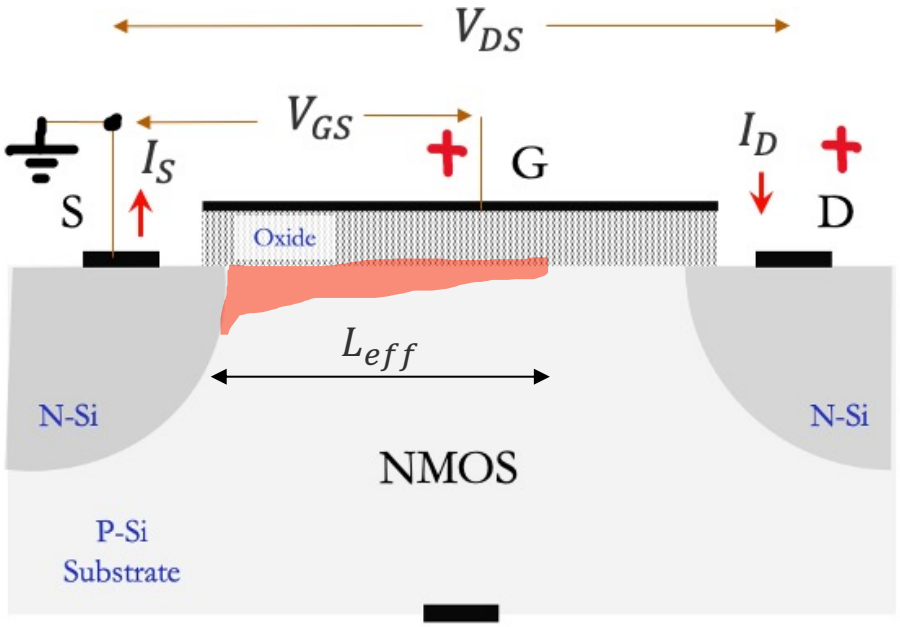
$\frac{W}{L} = \frac{400\,\mu m}{1\,\mu m}$

$k_n = 0.4\ \text{mA/V}^2$

$$R_S = \frac{-3 - (-5)}{0.4\,m} = 5\Omega k$$

$$R_D = \frac{5 - 1}{0.4\,m} = 10\Omega k$$

Channel Length Modulation

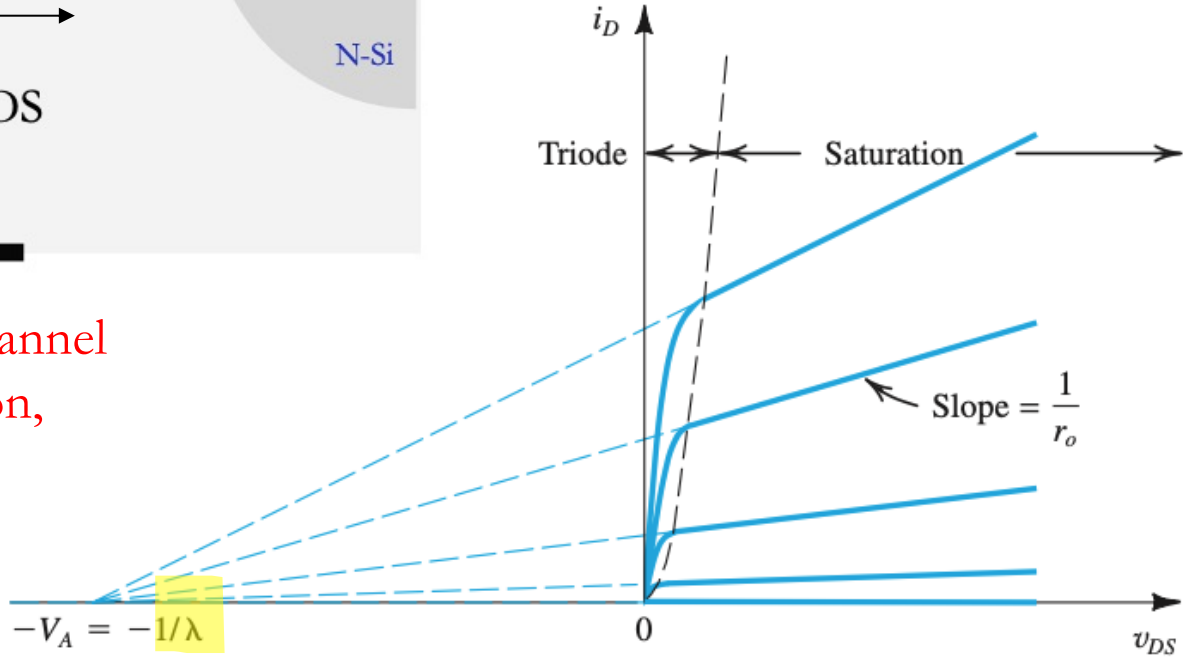


Saturation Mode

$$I_D = k(V_{GS} - V_t)^2(1 + \lambda V_{DS})$$

If we ignore channel length modulation,

$\lambda = 0$

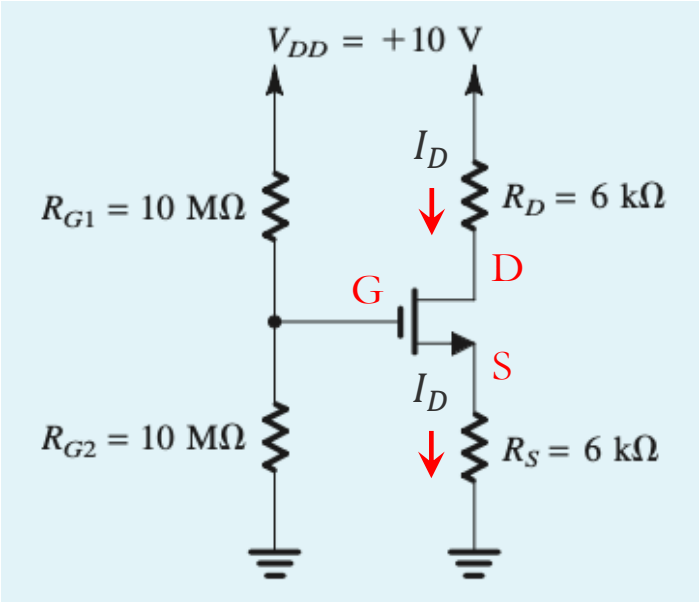


Summary

Mode	NMOS (V_t +ve)	PMOS (V_t -ve)
Cutoff	$V_{GS} < V_t \quad I_D = 0$	$V_{GS} > V_t \quad I_D = 0$
Triode	$V_{GS} \geq V_t$ and $V_{DS} \leq V_{GS} - V_t$ $I_D = k\left[2(V_{GS} - V_t)V_{DS} - {V_{DS}}^2\right]$	$V_{GS} \leq V_t$ and $V_{DS} \geq V_{GS} - V_t$ $I_D = k\left[2(V_{GS} - V_t)V_{DS} - {V_{DS}}^2\right]$
Saturation	$V_{GS} \geq V_t$ and $V_{DS} \geq V_{GS} - V_t$ $I_D = k(V_{GS} - V_t)^2(1 + \lambda V_{DS})$	$V_{GS} \leq V_t$ and $V_{DS} \leq V_{GS} - V_t$ $I_D = k(V_{GS} - V_t)^2(1 + \lambda V_{DS})$

Examples

5) Analyze the transistor circuit below. Take $V_t = 1\,V, k_n = 0.5$ mA/V^2.



Check mode of operation

$V_G = 5\,V$

V_{DS} ? $V_{GS} - V_t$

Can't decide!

Make an assumption.

Assume Saturation Mode

$V_D = 10 - (6\,k)I_D$ $\qquad$ $V_S = (6\,k)I_D$

$V_{DS} = 10 - (12\,k)I_D$ $\qquad$ $V_{GS} = 5 - (6\,k)I_D$

$$I_D = k(V_{GS} - V_t)^2$$

$$I_D = 0.5\,m(5 - (6\,k)I_D - 1)^2$$

$$18\,I_D^2 - 25I_D + 8 = 0$$

~~$I_D = 0.89\,mA$~~ $\qquad$ $I_D = 0.5\,mA$ ✓

~~$V_{GS} = -0.34\,V$~~ $\qquad$ $V_{GS} = +2\,V$

Cutoff

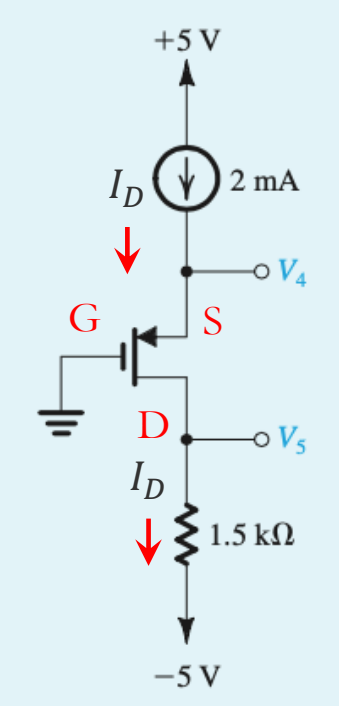
Check assumption

V_{DS} ? $V_{GS} - V_t$

$4 > 2 - 1$ ✓

Examples

6) Analyze the transistor circuit below. Take $|V_t| = 2\ V, k_p = 0.5\ \text{mA/V}^2, \lambda = 0.$



$$I_D = k(V_{GS} - V_t)^2$$

$$2\ m = 0.5\ m(V_{GS} + 2\ )^2$$

$$V_{GS} + 2 = \pm 2$$

$$V_{GS} + 2 = -2$$

$$V_{GS} = -4\ V$$

$$V_S = 4\ V$$

Check mode of operation

$V_G = 0 \qquad I_D = 2\ mA \qquad V_D = -2\ V$

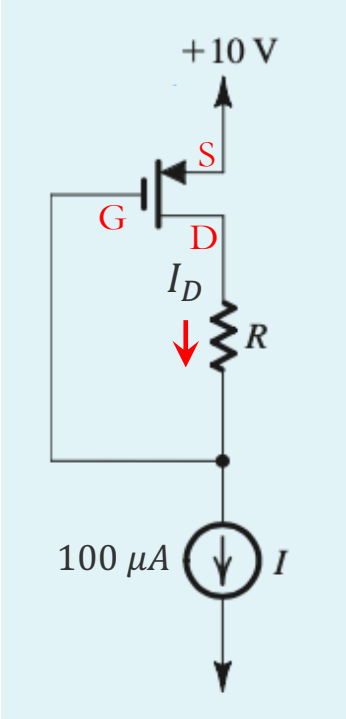
$V_D \quad ? \quad V_G - V_t$

$-2 \quad < \quad 0 + 2$

Saturation Mode

Examples

7) Analyze the transistor circuit below. Take $|V_t| = 1\ V, k_p = 0.1\ \text{mA/V}^2, \lambda = 0.$



Check mode of operation

$V_S = 10\ V \quad I_D = 0.1\ mA$

$V_{DS} \quad ? \quad V_{GS} - V_t$

$V_D \quad ? \quad V_G + 1$

It Depends on R

a) $R = 10\ k\Omega$

$$V_D - V_G = I_D R = (0.1\ m)(10\ k) = 1\ V$$ Saturation

$$I_D = k(V_{GS} - V_t)^2$$

$$0.1\ m = 0.1\ m(V_{GS} + 1)^2$$

$$V_{GS} = -2\ V \qquad V_{DS} = -1\ V$$

b) $R = 100\ k\Omega$

$$V_D - V_G = I_D R = (0.1\ m)(100\ k) = 10\ V$$ Triode

$$I_D = k\left[2(V_{GS} - V_t)V_{DS} - {V_{DS}}^2\right]$$

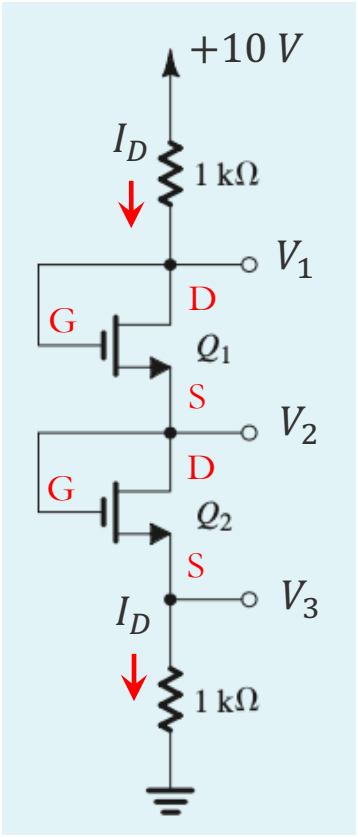
$$0.1\ m = 0.1\ m\left[2(V_{GS} + 1)(V_{GS} + 10) - (V_{GS} + 10)^2\right]$$

$$V_{GS} = -10.1\ V \qquad V_{DS} = -0.1\ V$$

$$V_{DS} = V_{DG} + V_{GS}$$

Examples

8) Analyze the transistor circuit below. Take $|V_t| = 1\ V, k_n = 1\ \text{mA/V}^2, \lambda = 0$.



$$V_3 = (V_2 - V_3 - 1)^2 \quad (3)$$

$$I_D = \frac{10 - V_1}{1\ k} = \frac{V_3}{1\ k} = (1\ m)(V_1 - V_2 - 1)^2 = (1\ m)(V_2 - V_3 - 1)^2$$

$$V_3 = 10 - V_1 \quad (1) \qquad V_3 = (V_1 - V_2 - 1)^2 \quad (2)$$

(4)

From 2 and 3 $\quad V_1 - V_2 - 1 = V_2 - V_3 - 1 \quad \rightarrow \quad V_3 = 2V_2 - V_1$

From 1 and 4 $\quad 10 - V_1 = 2V_2 - V_1 \quad \rightarrow \quad V_2 = 5\ V$

From 3 $\quad V_3 = (5 - V_3 - 1)^2 \quad \rightarrow \quad V_3 = 2.45\ V$

~~$V_3 = 6.55\ V$~~

Check mode of operation

Both transistors in saturation

$V_D = V_G$

$$I_D = \frac{V_3}{1k} = 2.45\ mA \qquad V_1 = 10 - 2.45 = 7.55\ V$$

Q & A

Nano Photonics and Plasmonics Research

Send me an email if
you have any
questions.